

CLICK Bench Meter Wiring

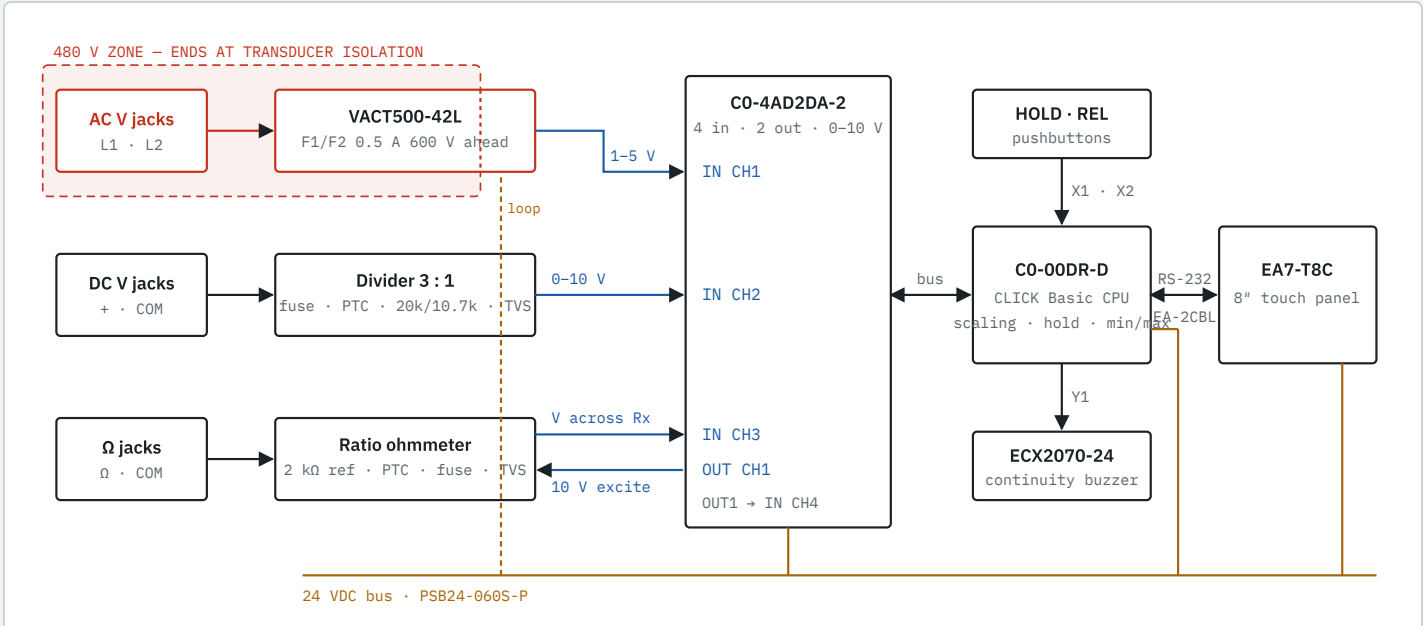
A three-function bench meter (AC volts to 480 V, DC volts to 30 V, ohms and continuity) built from a CLICK Basic PLC, one analog combo module and a C-more EA7-T8C touch panel.

CPU **C0-00DR-D** Analog **C0-4AD2DA-2** HMI **EA7-T8C** AC transducer **VACT500-42L**

Supply **PSB24-060S-P**

— line voltage (up to 480 VAC) — analog signal (0–10 V) — 24 VDC power — return / 0 V

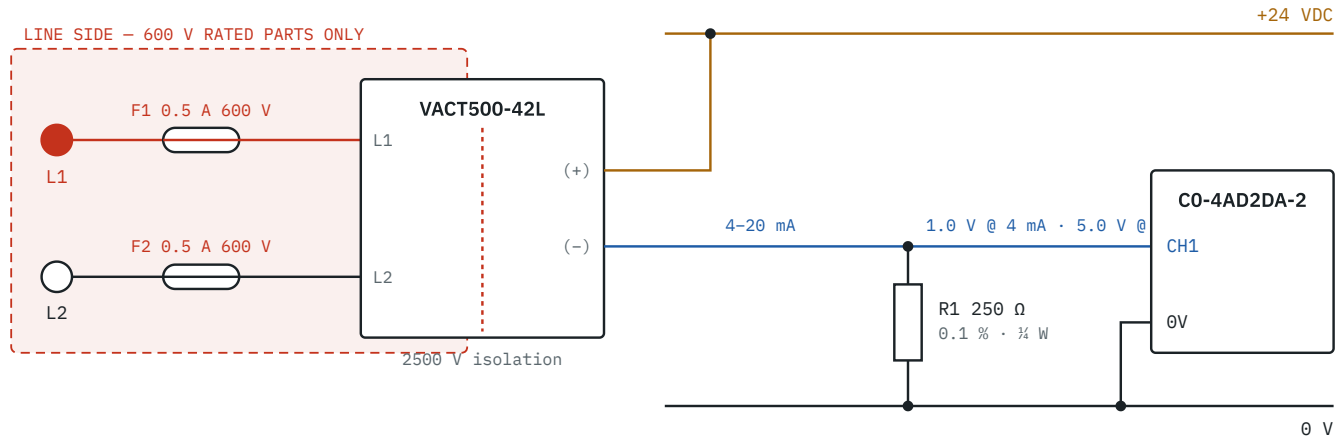
System overview



Each function has its own jacks and its own path to its own module input. Line voltage only reaches the VACT500-42L. Its isolated 4–20 mA output crosses into the low-voltage side. The DC and Ω circuits never share a conductor with the AC jacks, so choosing a mode on the touchscreen can't route 480 V anywhere.

AC volts – module input CH1

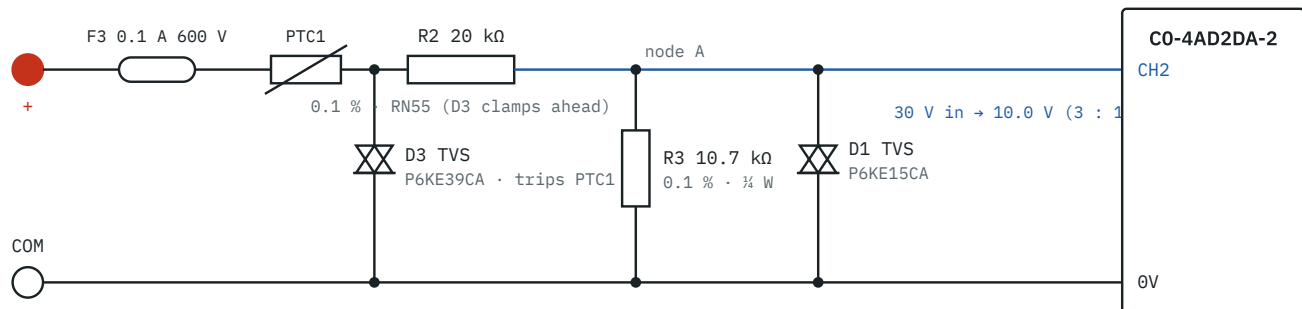
UP TO 480 VAC



The VACT is a 2-wire loop device. +24 V goes into its (+) terminal, and current leaves (-) and flows through R1 to 0 V. CH1 reads the voltage across R1. Use 250 Ω: the datasheet caps the load at 500 Ω, and 250 Ω leaves the transducer plenty of voltage. The output caps at 24 mA (6.0 V), so anything over about 520 V displays as OL. The 0.5 A 600 V fast-acting fuses are the value the VACT datasheet calls for.

DC volts – module input CH2

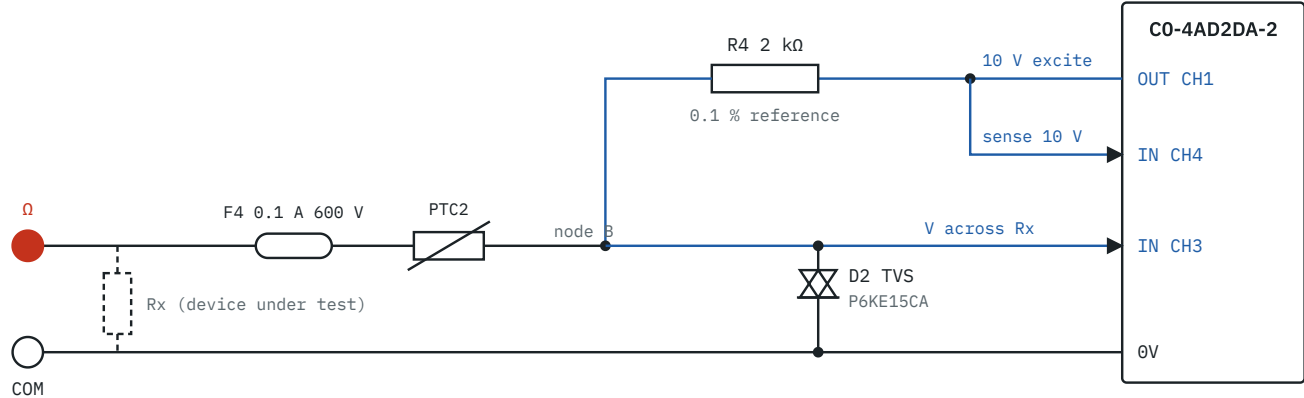
0-30 VDC



A true 3:1 divider: 30 V in gives 10 V at CH2. R3 is 10.7 kΩ instead of 10 kΩ because the module's input impedance (about 150 kΩ) sits in parallel with it: $10.7\text{ k}\Omega \parallel 150\text{ k}\Omega \approx 9.99\text{ k}\Omega$, which makes the ratio 3.002:1. The spec sheet only guarantees "over 150 kΩ", so check the ratio once against a known meter and trim the $\times 3.000$ factor in the ladder if it's off. **Fault protection:** if someone puts 480 V on these jacks, D3 clamps right behind PTC1 and pulls amps, so PTC1 trips (or F3 blows) in milliseconds. Without D3, R2 would limit the fault to about 24 mA, too little to trip anything, and R2 would cook at about 11 W. D1 is a second clamp right at the module. This is protection, not a rating: the input is 30 V.

Ohms & continuity – CH3, CH4, OUT CH1

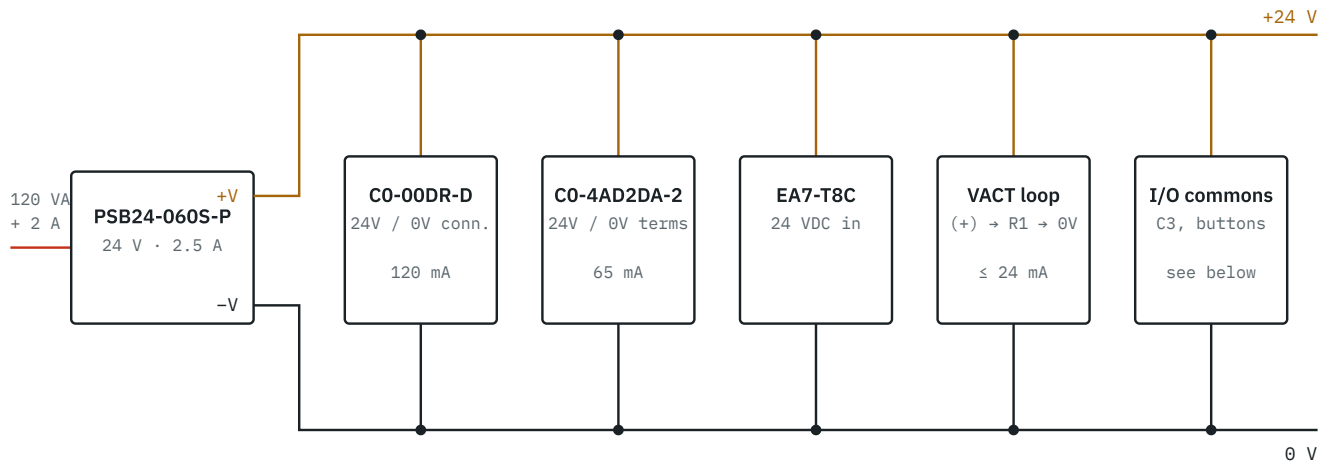
≈1 Ω – 200 KΩ



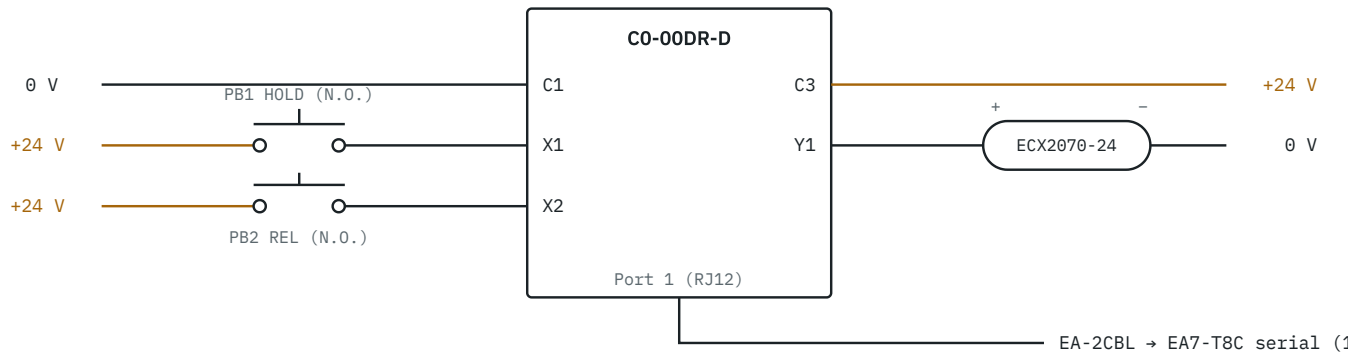
This is a ratio ohmmeter. OUT CH1 drives 10 V through R4 into the unknown, and CH4 measures the real excitation. The math uses the ratio of CH3 to CH4, which cancels the output's $\pm 1\%$ error. Short-circuit current is 5 mA, well under the output's 10 mA limit. PTC2, F4 and the leads add a few ohms, so short the leads and press REL to null them.

Power, buttons, buzzer, panel link

24 VDC



Everything runs off one 24 VDC supply. The total draw is well under 1 A including the panel, so the 2.5 A PSB24 has plenty of headroom. The VACT datasheet calls for a **Class 2** supply. Check the PSB24-060S-P's listing, and if it isn't Class 2, put a 4 A fuse in the VACT's +24 V lead (Class 2 limits at 24 V are about 100 VA / 4 A).



Inputs X1–X4 share C1. Tie C1 to 0 V and the buttons switch +24 V in. **Relay outputs Y1–Y4 share C3.** Feed C3 +24 V and Y1 switches the buzzer. Port 1 is also the programming port: unplug the EA-2CBL when you download ladder changes. Port 1 defaults are 38400 baud, 8-O-1, node 1, Modbus RTU.

Point-to-point wire list

FROM	TO	NOTES
POWER		
PSB24 +V	CPU 24V · module 24V · T8C 24V · VACT (+) · CPU C3 · PB1/PB2 one side	+24 V rail, 18 AWG
PSB24 –V	CPU 0V · module 0V (power) · T8C 0V · R1 low end · CPU C1 · buzzer (–)	0 V rail
120 VAC cord	PSB24 L / N / $\perp$	2 A fuse in line, ground the enclosure
AC VOLTS – 600 V RATED WIRE & PARTS		
AC jack L1 (red)	F1 → VACT L1	F1 KLKD.500 (0.5 A 600 V) in EHM1DU
AC jack L2 (black)	F2 → VACT L2	F2 KLKD.500 (0.5 A 600 V) in EHM1DU
VACT (–)	module IN CH1 + R1 high end	R1 250 Ω 0.1 %
R1 low end	module IN 0V (next to CH1)	
DC VOLTS		
DC jack + (red)	F3 → PTC1 → D3 to 0 V → R2 20 k Ω → node A	D3 sits between PTC1 and R2
node A	module IN CH2 · R3 10.7 k Ω to 0 V · D1 to 0 V	

FROM	TO	NOTES
DC jack COM (black)	module IN 0V (next to CH2)	
OHMS & CONTINUITY		
module OUT CH1	R4 2 k Ω → node B · jumper to IN CH4	CH4 senses excitation
node B	module IN CH3 · D2 to 0 V · PTC2 → F4 → Ω jack	
Ω jack COM (black)	module IN 0V (next to CH3)	
module OUT CH2	—	not connected
CONTROLS & COMMS		
+24 V → PB1 HOLD	CPU X1	N.O. pushbutton
+24 V → PB2 REL	CPU X2	N.O. pushbutton
CPU Y1	ECX2070-24 (+)	buzzer (–) to 0 V
CPU Port 1	T8C serial port	EA-2CBL

Small parts and the scaling math

Digi-Key parts

R1	RN55E2500BB14 · 250 Ω 0.1 %
R2	RN55E2002BB14 · 20 k Ω 0.1 %
R3	RN55E1072BB14 · 10.7 k Ω 0.1 %
R4	RN55E2001BB14 · 2 k Ω 0.1 %
D1, D2	P6KE15CA Littelfuse · TVS, 12.8 V standoff
D3	P6KE39CA Littelfuse · TVS, 33 V standoff
PTC1, 2	TR600-150 Littelfuse · 150 mA hold, 250 V operating / 600 V interrupt
F3, F4	KLKD.100 Littelfuse · 0.1 A 600 V midget, in EHM1DU holders
F1, F2	KLKD.500 Littelfuse · 0.5 A 600 V midget
Jacks	72913-2 red ×3, 72913-0 black ×3 · Pomona, CAT III 1000 V

Ladder scaling (volts at each input)

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ACV = (V_CH1 - 1.000) × 125      → 0-  
      OL if ACV > 520
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DCV = V_CH2 × 3.000              (trim  
      OL if V_CH2 > 9.98
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 $\Omega$  = 2000 × V_CH3 / (V_CH4 - V_CH3)  
      OL if V_CH3 > 0.99 × V_CH4
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BEEP (Y1) =  $\Omega$  < 30
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Inputs are 13-bit (1.22 mV/count). Expect about 0.15 V/count and $\pm 1\%$ on AC (the VACT's rating), about 4 mV/count on DC, and the best ohms accuracy from 10 Ω to 50 k Ω . Drive OUT CH1 at full scale (10 V).

Before you power it up

- Only the AC jacks, F1/F2 and the VACT input are rated for 480 V. Use 600 V wire there and keep it physically apart from the low-voltage wiring.
- The DC and Ω COM jacks are tied to the 24 V 0 V rail. If you earth that rail, those black jacks are at earth, the same as a bench scope's ground clip.
- Test the AC channel first on 120 V against a known-good meter before you go anywhere near 480 V.
- Keep a CAT III/IV handheld for live troubleshooting. This meter is for the bench.